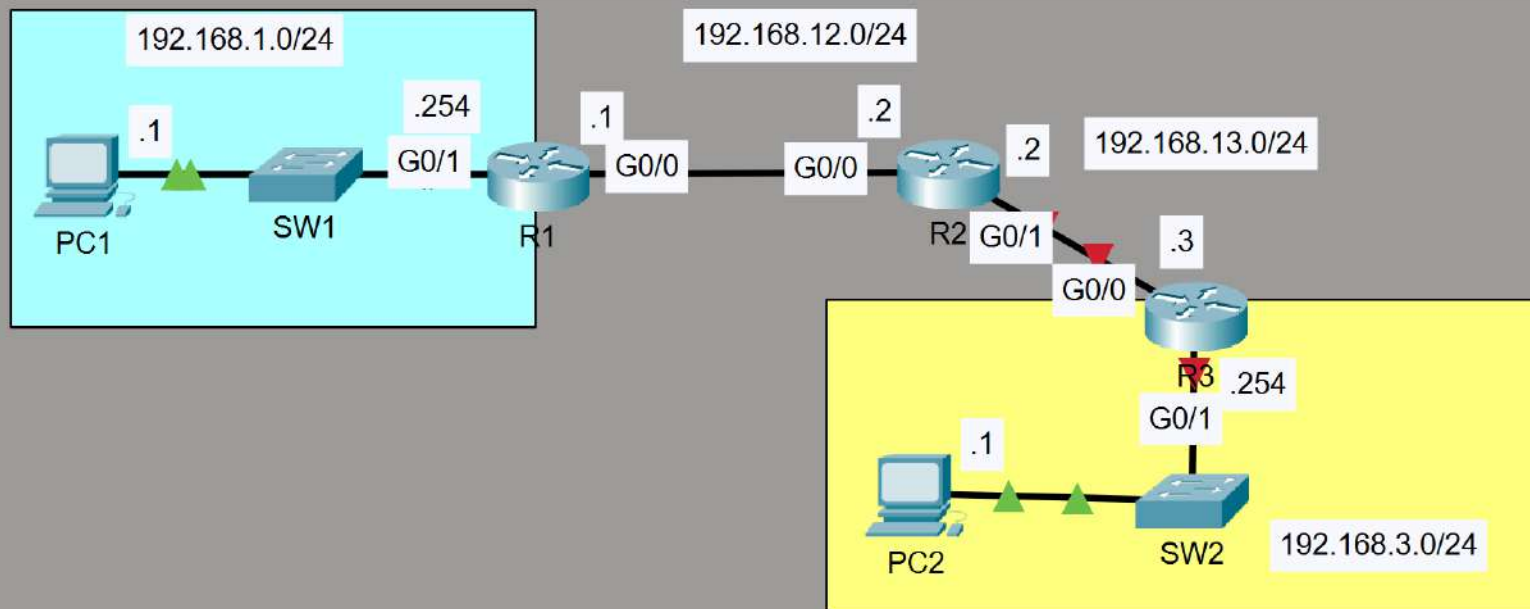


Today's Lab : Static Routing



Logical Physical x: 1062, y: 157

Root



Physical Config Desktop Programming Attributes**GLOBAL**

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ OnLink Speed ☐ 100 Mbps ☐ 10 Mbps ☒ AutoDuplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0001.C7D7.5066

IP Configuration

☐ DHCP☒ Static

IPv4 Address 192.168.12.1

Subnet Mask 255.255.255.0

IPv6 Configuration

☒ Automatic☒ Static

IPv6 Address

Link Local Address: FE80::201:C7FF:FED7:5066



PC1



Physical

Config

Desktop

Programming

Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Global Settings

Display Name

Interfaces

Gateway/DNS IPv4

☒ DHCP

☒ Static

Default Gateway

DNS Server

Gateway/DNS IPv6

☒ Automatic

I started configuring Router R1 by assigning an IP address to the interface connected to SW1. Then I enabled the interface with the no shutdown command

```
Router>
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface g0/1 192.168.1.254 255.255.255.0
                        ^
% Invalid input detected at '^' marker.

R1(config)#interface g0/1
R1(config-if)#ip add 192.168.1.254 255.255.255.0
R1(config-if)#des
R1(config-if)#description ##to SW1##
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed
state to up
```

```

R1(config-if)#
R1(config-if)#interf
R1(config-if)#interface g0/0
R1(config-if)#ip add 192.168.12.1 255.255.255.0
R1(config-if)#des
R1(config-if)#description ##TO R2##
R1(config-if)#do show ip int brief
Interface                IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0      192.168.12.1    YES manual  administratively down down
GigabitEthernet0/1      192.168.1.254   YES manual  up          up
GigabitEthernet0/2      unassigned      YES unset   administratively down down
Vlan1                   unassigned      YES unset   administratively down down
R1(config-if)#no shutdown

```

I configured R1's second interface facing R2 and verified the configuration using show ip interface brief. This confirmed the interface addresses and operational status.

```

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
do show ip int brief
Interface                IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0      192.168.12.1    YES manual  up          down
GigabitEthernet0/1      192.168.1.254   YES manual  up          up
GigabitEthernet0/2      unassigned      YES unset   administratively down down
Vlan1                   unassigned      YES unset   administratively down down
R1(config-if)#

```

3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

I configured Router R2 by assigning IP addresses to its interfaces connecting to R1 and R3. The interfaces were then enabled to establish connectivity between routers

```
Router>
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#interface g0/0
R2(config-if)#ip add 192.168.12.2
% Incomplete command.
R2(config-if)#ip add 192.168.12.2 255.255.255.0
R2(config-if)#des
R2(config-if)#description ##To R1##
R2(config-if)#no shut
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed
state to up

R2(config-if)#
R2(config-if)#interface g0/1
R2(config-if)#ip add 192.168.13.2 255.255.255.0
R2(config-if)#
```

I configured Router R3 with IP addresses on both the inter-router link and the LAN-facing interface. A verification command was used to confirm the interfaces were up and correctly addressed.

```
Router>
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R3
R3(config)#interface g0/0
R3(config-if)#ip add 192.168.13.3 255.255.255.0
R3(config-if)#des
R3(config-if)#description ##To R2###
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed
state to up

R3(config-if)#interface g0/1
R3(config-if)#ip add 192.168.3.0 255.255.255.0
Bad mask /24 for address 192.168.3.0
R3(config-if)#ip add 192.168.3.254 255.255.255.0
R3(config-if)#des
R3(config-if)#description #TO SW2#
R3(config-if)#no sh
R3(config-if)#no shutdown

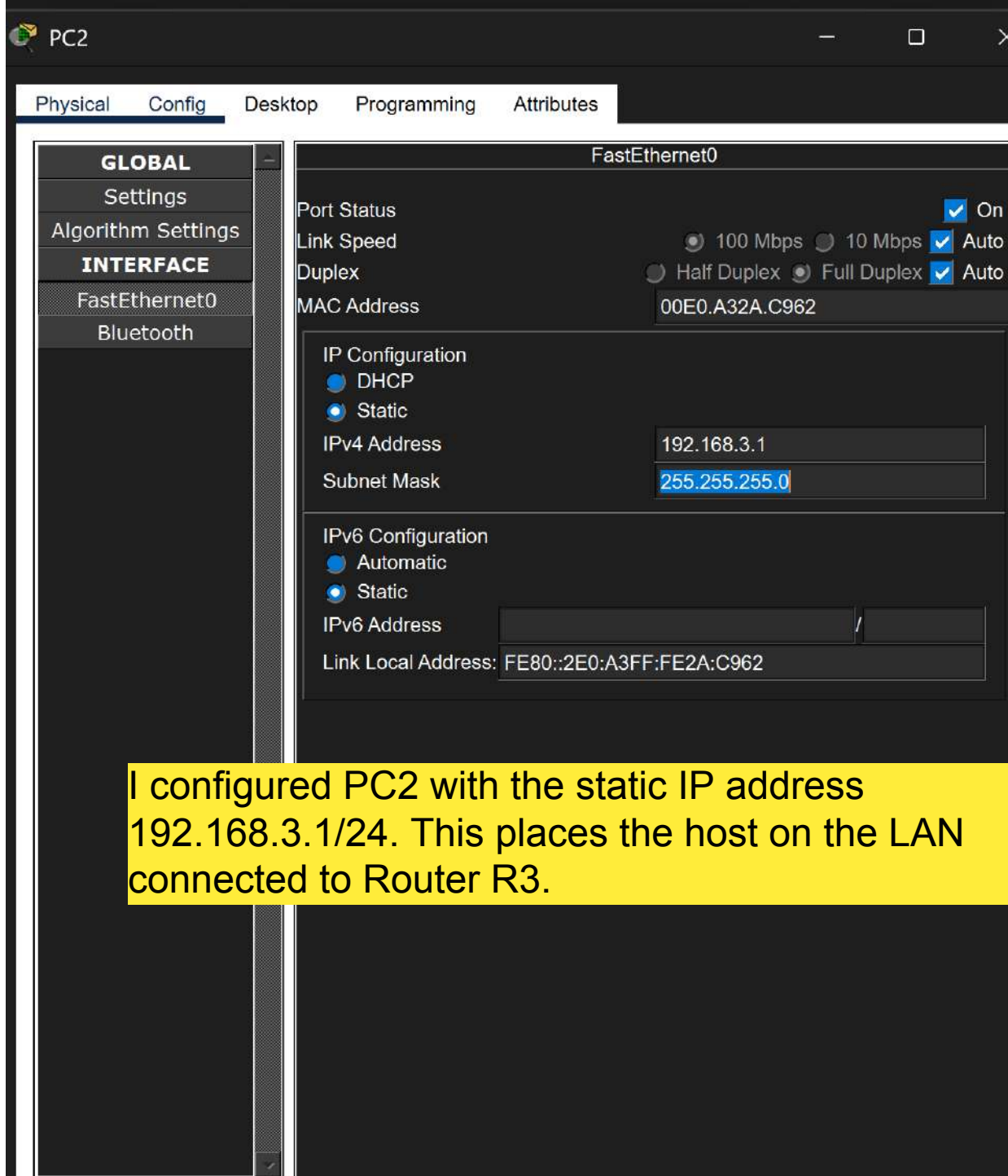
R3(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed
state to up
do show ip int brrief
show ip int brrief
^
% Invalid input detected at '^' marker.

R3(config-if)#do show ip int bri
R3(config-if)#do show ip int brief
Interface                IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0      192.168.13.3    YES manual up
```

Copy

Paste



I configured PC2 with the static IP address 192.168.3.1/24. This places the host on the LAN connected to Router R3.

PC2

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Global Settings

Display NamePC2

InterfacesFastEthernet0

Gateway/DNS IPv4

☐ DHCP

☒ Static

Default Gateway192.168.3.254

DNS Server

Gateway/DNS IPv6

☐ Automatic

☒ Static

Default Gateway

DNS Server

Device Clock:19:17:01 Sun Feb 28 1993 UTC

NTP

Server Ip:

Authentication

☐ Enable

☒ Disable

Password:

PTP

ProfilePower

Top

ab - Configuring Static Routes.pkt

R1

Physical Config CLI Attributes

Press RETURN to get started.

I began configuring static routes on Router R1 to reach the remote 192.168.3.0/24 network. The route points traffic toward the next-hop router, R2.

```
R1>
R1>enable
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route ?
  A.B.C.D  Destination prefix
R1(config)#ip route 192.168.3.0 ?
  A.B.C.D  Destination prefix mask
R1(config)#ip route 192.168.3.0 255.255.255.0 ?
  A.B.C.D      Forwarding router's address
  Dialer        Dialer interface
  Ethernet       IEEE 802.3
  FastEthernet   FastEthernet IEEE 802.3
  GigabitEthernet GigabitEthernet IEEE 802.3z
  Loopback       Loopback interface
  Null           Null interface
  Serial         Serial
  Vlan           Catalyst Vlans
R1(config)#ip route 192.168.3.0 255.255.255.0 192.168.12.2 ?
  <1-255>  Distance metric for this route
  <cr>
R1(config)#ip route 192.168.3.0 255.255.255.0 192.168.12.2
R1(config)#
```

Copy Paste

Top

I verified the static route configuration on R1 using show ip route. The routing table now includes the remote network learned through the manually configured static route

ab - Configuring Static Routes.pkt

R1

Physical Config CLI Attributes

```
R1>enable
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip route ?
  A.B.C.D Destination prefix
R1(config)#ip route 192.168.3.0 ?
  A.B.C.D Destination prefix mask
R1(config)#ip route 192.168.3.0 255.255.255.0 ?
  A.B.C.D Forwarding router's address
  Dialer Dialer interface
  Ethernet IEEE 802.3
  FastEthernet FastEthernet IEEE 802.3
  GigabitEthernet GigabitEthernet IEEE 802.3z
  Loopback Loopback interface
  Null Null interface
  Serial Serial
  Vlan Catalyst Vlans
R1(config)#ip route 192.168.3.0 255.255.255.0 192.168.12.2 ?
  <1-255> Distance metric for this route
  <cr>
R1(config)#ip route 192.168.3.0 255.255.255.0 192.168.12.2
R1(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
       area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/1
L       192.168.1.254/32 is directly connected, GigabitEthernet0/1
S       192.168.3.0/24 [1/0] via 192.168.12.2
    192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected, GigabitEthernet0/0
L       192.168.12.1/32 is directly connected, GigabitEthernet0/0

R1(config)#
```

Copy Paste

Top

I attempted to configure a static route on Router R2 using the outgoing interface as the next hop. Packet Tracer generated a warning because the link is not a point-to-point connection.

R2

Physical Config CLI Attributes

```
R2(config-if)#description ##To R1##
R2(config-if)#no shut
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed
state to up

R2(config-if)#
R2(config-if)#interface g0/1
R2(config-if)#ip add 192.168.13.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

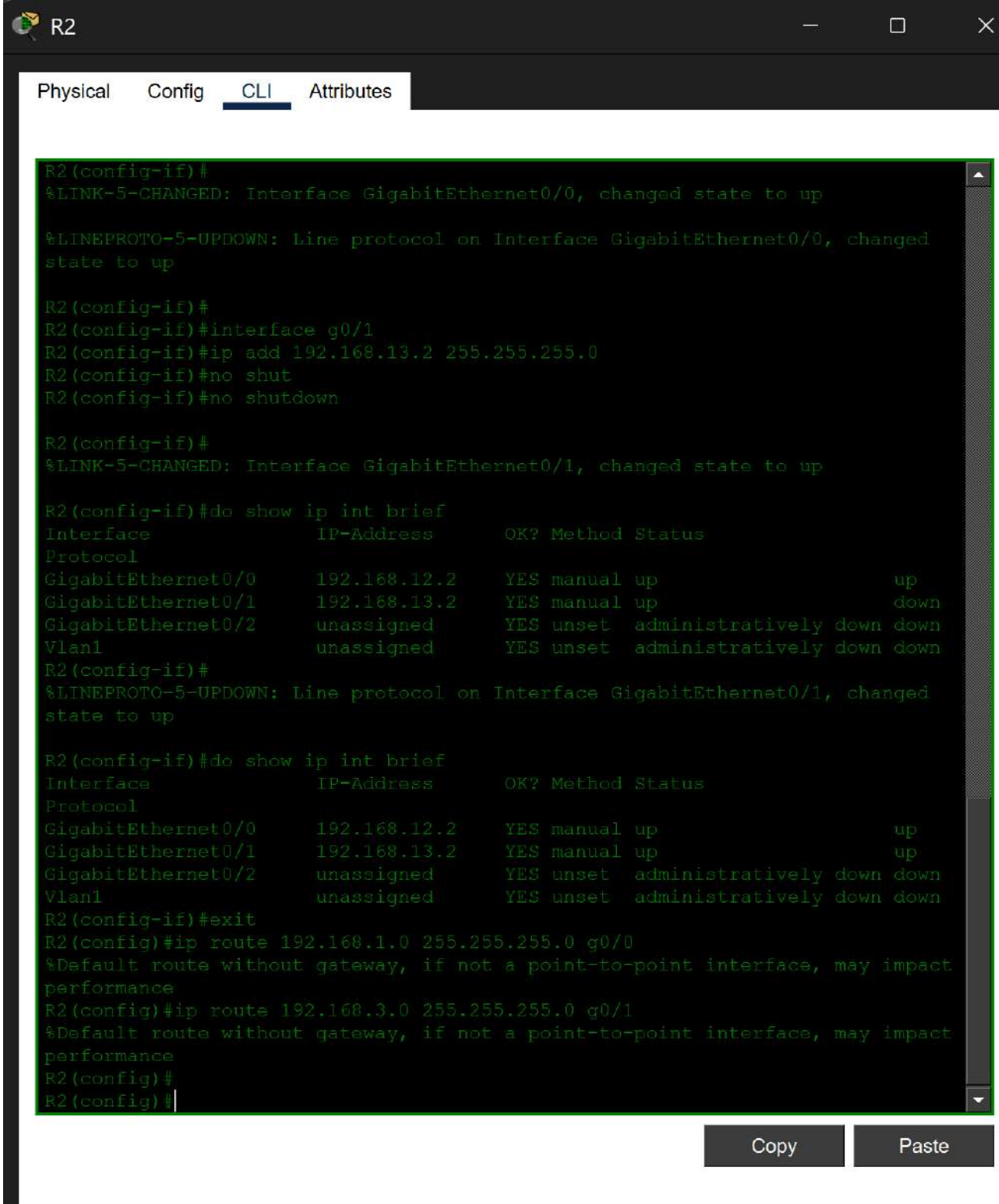
R2(config-if)#do show ip int brief
Interface                IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0       192.168.12.2    YES manual up
GigabitEthernet0/1       192.168.13.2    YES manual up
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
R2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed
state to up

R2(config-if)#do show ip int brief
Interface                IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0       192.168.12.2    YES manual up
GigabitEthernet0/1       192.168.13.2    YES manual up
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
R2(config-if)#exit
R2(config)#ip route 192.168.1.0 255.255.255.0 g0/0
%Default route without gateway, if not a point-to-point interface, may impact
performance
R2(config)#
```

Copy Paste

Top

I completed the static route configuration on R2 to provide connectivity between the networks on either side of the router. This allows R2 to forward traffic toward the correct destination network



```
R2
Physical Config CLI Attributes

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed
state to up

R2(config-if)#
R2(config-if)#interface g0/1
R2(config-if)#ip add 192.168.13.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

R2(config-if)#do show ip int brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     192.168.12.2    YES manual up
GigabitEthernet0/1     192.168.13.2    YES manual up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed
state to up

R2(config-if)#do show ip int brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     192.168.12.2    YES manual up
GigabitEthernet0/1     192.168.13.2    YES manual up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R2(config-if)#exit
R2(config)#ip route 192.168.1.0 255.255.255.0 g0/0
%Default route without gateway, if not a point-to-point interface, may impact
performance
R2(config)#ip route 192.168.3.0 255.255.255.0 g0/1
%Default route without gateway, if not a point-to-point interface, may impact
performance
R2(config)#
R2(config)#
```

Copy Paste

I configured and verified the static route on Router R3 using show ip route. The routing table confirms that the remote network is reachable through Router R2

```
R3
Physical Config CLI Attributes

Press RETURN to get started.

R3>
R3>enable
R3#conf t

R3(config)#ip route 192.168.1.0 255.255.255.0 192.168.13.2
R3(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
       area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    192.168.1.0/24 [1/0] via 192.168.13.2
     192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.3.0/24 is directly connected, GigabitEthernet0/1
L     192.168.3.254/32 is directly connected, GigabitEthernet0/1
     192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.13.0/24 is directly connected, GigabitEthernet0/0
L     192.168.13.3/32 is directly connected, GigabitEthernet0/0

R3(config)#
```

Copy

Paste

I performed an initial ping test from PC1 to PC2 before all routing was fully configured. I believe it failed because ARP request were taking too long.

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.3.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
```

Simulation Panel

Event List

Vis.	Time(sec)	Last Device
	1.682	SW2
	1.692	--
	1.693	SW1
	1.693	SW1
	3.682	--
Visible	3.683	SW2
Visible	3.683	SW2

Reset Simulation ☒ Constant Delay Captured to 3.683

Play Controls



C:\>

C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Reply from 192.168.3.1: bytes=32 time=11ms TTL=125

Reply from 192.168.3.1: bytes=32 time=1ms TTL=125

Reply from 192.168.3.1: bytes=32 time<1ms TTL=125

Reply from 192.168.3.1: bytes=32 time=64ms TTL=125

Ping statistics for 192.168.3.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 64ms, Average = 19ms

C:\>